

A novel RSSI-based geometric iterative centroid positioning optimization strategy using commercial Wi-Fi signals

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Abstract

The information interaction and positioning technology based on commercial Wi-Fi plays a crucial role in expanding the applications for the Internet of Things and smart homes. Achieving high-precision indoor positioning through the optimal deployment of Wi-Fi access points remains a significant challenge. This study proposes two indoor positioning optimization algorithms based on the received signal strength indicator (RSSI) of commercial Wi-Fi signals. The enhanced weighted centroid positioning (EWCP) algorithm introduces a novel weighting method that effectively leverages RSSI ranging errors to improve positioning accuracy and further minimize positioning errors. Meanwhile, the adaptive iterative weighted centroid positioning (AIWCP) algorithm incorporates a detailed weighting approach during the movement of the signal transmitter, providing specific movement indications and enhancing positioning performance. Both simulation and experimental results confirm the effectiveness of these two positioning methods.

KEYWORDS

IoT, localization, RSSI, Wi-Fi

1 | INTRODUCTION

With the rapid advancement of wireless communication technology [1], wireless sensor technology has found widespread application [2,3], playing a unique and irreplaceable role in various fields, such as industrial production, military reconnaissance, and positioning systems [4–6]. Positioning systems are primarily categorized into outdoor and indoor applications. Traditional outdoor positioning systems based on global positioning systems (GPS) exhibit poor performance in complex indoor environments [7] because of signal obstructions caused by roofs and building interiors, resulting in significant inaccuracies [8]. To address these limitations, indoor positioning technology was developed as an extension of

satellite positioning. In complex indoor environments, this technology enables rapid localization of trapped individuals during rescue operations [9,10]. Indoor positioning algorithms are broadly classified into distance-free and distance-based algorithms, depending on whether they require distance information between the signal transmitter and receiver. Typical distance-based positioning algorithms include time of arrival (TOA) [11], time difference of arrival (TDOA) [12], and received signal strength indicator (RSSI) [13]. Compared with other positioning technologies [14], RSSI-based algorithms offer advantages such as low power consumption [15] and the elimination of the need for additional hardware [16], making them one of the most commonly used indoor positioning solutions [17].

To address the deficiencies in the accuracy improvement of traditional indoor positioning algorithms, this paper proposes two optimized indoor positioning algorithms based on Wi-Fi signal RSSI: the enhanced weighted centroid positioning (EWCP) algorithm and an adaptive iterative weighted centroid positioning (AIWCP) algorithm. These two optimization algorithms combine the characteristics of RSSI with the geometric relationships inherent in positioning scenarios. The main contributions of this study include the following:

- The designed EWCP algorithm effectively exploits the errors in RSSI ranging. Under the constraint of a limited number of signal transmitters, the algorithm maximizes the utilization of available positioning resources to improve accuracy.
- The proposed EWCP algorithm introduces a more effective weighting formula, achieving higher positioning accuracy compared to traditional weighted formulas. The algorithm fully leverages the characteristics of RSSI and the geometric relationships within the positioning scenario. Additionally, it incorporates an innovative method for calculating the intersection points of three circles using angle relationships, which significantly reduces computational complexity.
- The presented AIWCP algorithm builds upon and optimizes the weighted centroid positioning (WCP) algorithm. When the Wi-Fi signal transmitters are mobile, the algorithm can compute the coordinates that the transmitters need to reach. After the transmitters are repositioned to the specified coordinates, the algorithm ensures improved positioning accuracy is compared to the initial state.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 presents the logarithmic path loss model and traditional positioning algorithms. The proposed EWCP algorithm and AIWCP algorithm are introduced in Section 4. Simulation results are presented and analyzed in Section 5. Finally, Section 6 presents the conclusions of this study.

2 | RELATED WORKS

With the rapid advances in wireless sensor technology, research on RSSI-based indoor positioning algorithms continue to evolve. For instance, an improved centroid-localization algorithm utilizing virtual beacon nodes was proposed in [18]. To reduce energy consumption in positioning scenarios, a new indoor positioning system was proposed in [19]. This system employs a k-means clustering algorithm to better characterize indoor environments

and further refines location estimation using Bayesian inference.

Reference [20] introduces an interval random analysis approach to address signal instability in Wi-Fi indoor localization. [21] proposes combining particle swarm optimization with signal propagation models to optimize position estimation without prior training or parameter knowledge.

In [22], a weighted centroid localization algorithm based on the RSSI was introduced. This algorithm incorporates a three-dimensional Z-axis concept into the original method, thereby significantly enhancing accuracy. In addition, [23] proposed a triangular centroid positioning method coupled with a k-means clustering algorithm. This approach was specifically applied to scenarios involving the detection and tracking of counterfeit access points.

A hybrid indoor positioning algorithm was introduced in [24], which leveraging an inertial measurement unit tailored for pedestrians with swinging arms. Furthermore, [25] presented an indoor positioning algorithm featuring a rotating arc triangle layout. This method employs circular arc triangles to deploy beacon nodes and rotates their positions to enhance the average accuracy of positioning within the triangle. In [26], an enhanced centroid algorithm combined with the maximum likelihood estimation was proposed to improve the centroid algorithm for precise node coordinate estimation. Moreover, [27] introduced a filtered RSSI and beacon weight approach that utilizes a Kalman filter with improved RSSI measurements. The experimental results demonstrate high localization accuracy achieved through a cost-effective solution.

Indoor positioning technology encompasses numerous applications. [28] introduced a device-free human detection and tracking system using RSSI in indoor environments. This system integrates wireless communication with human-detection capabilities to effectively monitor and track human movements. Similarly, [29] focused on localization using LoRa technology, developing RSSI-based localization algorithms tailored for LoRa networks. This study introduced various algorithms designed to mitigate Gaussian and nonGaussian noise in LoRa environments.

3 | SYSTEM MODEL

3.1 | The logarithmic path loss model

By utilizing RSSI, the Wi-Fi signal strength at the receiver is determined. In indoor positioning, signal loss varies with the transmitter's location, resulting in different signal strengths. This power difference is used to calculate the distance between the transmitter and receiver. RSSI

ranging relies on the logarithmic path loss model, expressed as follows:

$$PL(q) = PL(q_0) + 10n \cdot \log_{10} \left(\frac{q}{q_0} \right) + N_0, \quad (1)$$

where q is the distance between the transmitter and receiver and q_0 denotes the reference distance to the transmitter. $PL(q)$ represents the signal loss at a distance q from the transmitter. $PL(q_0)$ represents the signal loss at a distance q_0 from the signal transmitter. n indicates the path loss coefficient. N_0 denotes the measurement noise, which assumes a Gaussian random variable with a mean value of zero and variance of σ^2 . According to the logarithmic distance loss model, the RSSI value is calculated as follows:

$$R_s = P_t - PL(q_0) - 10n \cdot \log_{10} \left(\frac{q}{q_0} \right) - N_0, \quad (2)$$

where P_t and R_s represent the transmission power and RSSI value, respectively. Based on (2), the distance q between the transmitter and receiver is calculated as $q = q_0 \times 10^{(P_t - PL(q_0) - R_s) \times (10n)^{-1}}$.

3.2 | Indoor trilateral positioning algorithm

Assuming that the positions of the transmitters are known and the position of the receiver is unknown, the receiver is defined as the target S whereas the transmitters are defined as locator nodes. Locator nodes extract the RSSI values from the target to compute the distances q between the target and each locator node. The indoor trilateral positioning (ITP) [30] algorithm utilizes the geometric structure of the locator nodes and target S for localization.

As shown in Figure 1, points D, E , and F indicate the locator nodes, with three circles centered on these nodes intersecting at point S . The coordinates of the three locator nodes are set as (x_1^a, y_1^b) , (x_2^a, y_2^b) , and (x_3^a, y_3^b) , respectively. The coordinates of the target S is set as (x_a, y_b) . The measured distances from D, E , and F to S are q_1, q_2 , and q_3 , respectively. These distances are derived from RSSI values and not from actual distances. The specific equations are as follows:

$$\begin{aligned} (x_a - x_1^a)^2 + (y_b - y_1^b)^2 &= q_1^2, \\ (x_a - x_2^a)^2 + (y_b - y_2^b)^2 &= q_2^2, \\ (x_a - x_3^a)^2 + (y_b - y_3^b)^2 &= q_3^2, \end{aligned} \quad (3)$$

where $q_1 = q_0 \times 10^{(P_t - PL(q_0) - R_{s1}) \times (10n)^{-1}}$, $q_2 = q_0 \times 10^{(P_t - PL(q_0) - R_{s2}) \times (10n)^{-1}}$, and $q_3 = q_0 \times 10^{(P_t - PL(q_0) - R_{s3}) \times (10n)^{-1}}$. Solving (3) yields the target position. As q_i denotes the measured distance, the circles often do not intersect at a single point. Thus, an analytical solution is not feasible. Consequently, a more sophisticated localization algorithm is required to improve positioning performance.

3.3 | The indoor centroid positioning algorithm

The conventional centroid positioning (CCP) [31] algorithm is range free and does not use RSSI. The WCP [32] algorithm utilizes RSSI to measure distances and employs these distances as weighted coefficients. As shown in Figure 2A, points D, E , and F are locator nodes with coordinates (x_1^a, y_1^b) , (x_2^a, y_2^b) , and (x_3^a, y_3^b) . The point $S(x_a, y_b)$ indicates the target.

The measured distances q_1, q_2 , and q_3 are introduced as weight coefficients to estimate the coordinates of the target. Multidimensional WCP (MWCP) [33] and single-dimensional WCP (SWCP) [34] algorithms can localize the target within a certain margin of error. The MWCP algorithm's positioning process is detailed in (4) and (5), whereas that of the SWCP algorithm is shown in (6) and (7).

$$x_a = \frac{x_1^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}} + x_2^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + x_3^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}}}{10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}}}, \quad (4)$$

$$y_b = \frac{y_1^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}} + y_2^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + y_3^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}}}{10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}}}. \quad (5)$$

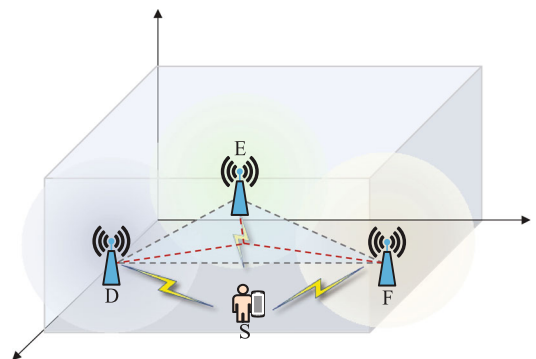


FIGURE 1 Schematic diagram of the ITP algorithm.

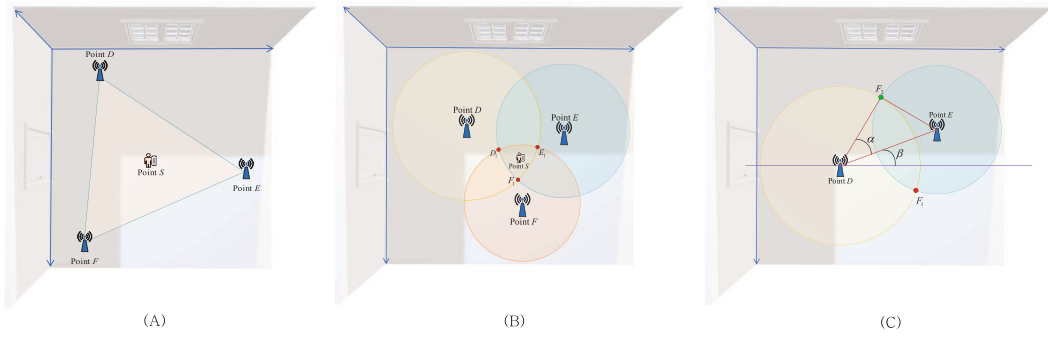


FIGURE 2 Schematic diagram of (A) the WCP algorithm, (B) the EWCP algorithm, and (C) the geometric angle method.

4 | OPTIMIZATION INDOOR POSITIONING ALGORITHMS

This paper introduces the EWCP algorithm, which utilizes geometric angle relationships to find intersection points, and the AIWCP algorithm, which incorporates weighted iterations to further optimize EWCP.

Under the same conditions as traditional algorithms, such as ITP and WCP, the EWCP algorithm requires only three locator nodes. Its core functionality lies in accurately determining the intersections formed by the three locator nodes. By effectively integrating the coordinate information of these intersections and applying a newly designed weighting method, the positioning accuracy is significantly improved. Furthermore, the AIWCP algorithm requires only three locator nodes and is specifically optimized for scenarios in which the locator nodes are movable. The main advantage of the AIWCP algorithm is that it can further enhance positioning accuracy each time the mobile locator nodes move to the designated coordinates. This method improves positioning performance in dynamic environments and provides more stable and reliable positioning results without the need for additional locator nodes.

4.1 | Enhanced weighted centroid positioning algorithm

This subsection describes the implementation of the EWCP algorithm. First, the proposed geometric angle method is employed to determine the intersection points resulting from RSSI ranging errors. Then, based on the coordinates of these intersection points, a newly designed weighting formula is utilized to accurately calculate the target's position.

During the localization process, the three locator nodes form circles that do not intersect precisely at the target point but instead intersect within an area. Thus, the coordinates of the three intersection points of these intersecting circles can be calculated. As shown in Figure 2B, the coordinates of the three locator nodes D, E , and F are set as (x_1^a, y_1^b) , (x_2^a, y_2^b) , and (x_3^a, y_3^b) , respectively. The measured distances between the three locator nodes D, E , and F and target S are set as q_1, q_2 , and q_3 , respectively. The locator nodes serve as the centers of the three circles, with the measured distances corresponding to their radii. The coordinates of the three intersections $D_1(x_{11}^a, x_{11}^b)$, $E_1(x_{22}^a, x_{22}^b)$, and $F_1(x_{33}^a, x_{33}^b)$ are unknown and need to be determined, as shown in Figure 2B.

$$x_a = \frac{x_1^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s2} + P_t - PL(q_0) - R_{s3}}{10n}} + x_2^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s3}}{10n}} + x_3^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s2}}{10n}}}{10^{\frac{P_t - PL(q_0) - R_{s2} + P_t - PL(q_0) - R_{s3}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s3}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s2}}{10n}}}, \quad (6)$$

$$y_b = \frac{y_1^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s2} + P_t - PL(q_0) - R_{s3}}{10n}} + y_2^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s3}}{10n}} + y_3^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s2}}{10n}}}{10^{\frac{P_t - PL(q_0) - R_{s2} + P_t - PL(q_0) - R_{s3}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s3}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s1} + P_t - PL(q_0) - R_{s2}}{10n}}}. \quad (7)$$

In this study, a geometric angle method was designed to determine the coordinates of the three intersections. As shown in Figure 2C, in the case of locator nodes D and E , the two circles intersect at points F_1 and F_2 . The length of the line segment $\overline{DF_2}$ is the radius q_1 of circle D , and the length of the line segment $\overline{EF_2}$ is the radius q_2 of circle E . The length of the line segment $\overline{DE}$ is represented by the symbol d_{DE} . Thus, the lengths of the three sides of the triangle $\triangle DEF_2$ are known.

The magnitude of angle $\angle EDF_2$ is represented by the symbol $\alpha_{R_{s1}}$. Then, the value of $\alpha_{R_{s1}}$ can be determined according to the law of cosines, as follows:

$$\begin{aligned}\alpha_{R_{s1}} &= \arccos \left[\frac{q_1^2 - q_2^2 + d_{DE}^2}{2 \times q_1 \times d_{DE}} \right] \\ &= \arccos \left\{ \frac{q_0^2 \left[10^{\frac{P_t - PL(q_0) - R_{s1}}{5n}} - 10^{\frac{P_t - PL(q_0) - R_{s2}}{5n}} \right] + d_{DE}^2}{2 \times d_{DE} \times q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}}} \right\}.\end{aligned}\quad (8)$$

According to the intersection properties of the circles, the magnitude of angle $\angle EDF_2$ is equal to that of angle $\angle EDF_1$. In Figure 2C, the purple line passing through the two circles is parallel to the x-axis. Consequently, the angle between the line segment $\overline{DE}$ and the yellow line is the same as the angle between the line segment $\overline{DE}$ and the x-axis. The magnitude of this angle is represented by the symbol $\beta_{R_{s1}}$. The formula for calculating $\beta_{R_{s1}}$ is expressed as $\beta_{R_{s1}} = \arctan \left((y_2^b - y_1^b) \times (x_2^a - x_1^a)^{-1} \right)$.

It should be noted that when the horizontal coordinates of locator nodes D and E are the same, the line segment $\overline{DE}$ is perpendicular to the x-axis. At this time, the above formula cannot be used to calculate the value of $\beta_{R_{s1}}$, and $\beta_{R_{s1}}$ should be directly assigned to 90° . Finally, the coordinates of the two intersection points, $F_2(x_{32}^a, y_{32}^b)$ and $F_1(x_{33}^a, y_{33}^b)$, can be calculated as shown in (9)–(16).

$$\begin{aligned}x_{32}^a &= x_1^a + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \cos(\beta_{R_{s1}} + \alpha_{R_{s1}}) \\ &= x_1^a + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \\ &\quad \times \left\{ \cos \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \cdot \frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right. \\ &\quad \left. - \sin \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \right. \\ &\quad \left. \cdot \sin \left[\arccos \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right) \right] \right\},\end{aligned}\quad (9)$$

$$\begin{aligned}y_{32}^b &= y_1^b + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \sin(\beta_{R_{s1}} + \alpha_{R_{s1}}) \\ &= y_1^b + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \\ &\quad \times \left\{ \sin \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \cdot \frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right. \\ &\quad \left. + \cos \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \right. \\ &\quad \left. \cdot \sin \left[\arccos \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right) \right] \right\},\end{aligned}\quad (10)$$

$$\begin{aligned}x_{33}^a &= x_1^a + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \cos(\beta_{R_{s1}} - \alpha_{R_{s1}}) \\ &= x_1^a + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \\ &\quad \times \left\{ \cos \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \cdot \frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right. \\ &\quad \left. + \sin \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \right. \\ &\quad \left. \cdot \sin \left[\arccos \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right) \right] \right\},\end{aligned}\quad (11)$$

$$\begin{aligned}y_{33}^b &= y_1^b + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \sin(\beta_{R_{s1}} - \alpha_{R_{s1}}) \\ &= y_1^b + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \\ &\quad \times \left\{ \sin \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \cdot \frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right. \\ &\quad \left. - \cos \left[\arctan \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right) \right] \right. \\ &\quad \left. \cdot \sin \left[\arccos \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right) \right] \right\}.\end{aligned}\quad (12)$$

$$\begin{aligned}x_{32}^a &= x_1^a + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \left\{ \frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE} \sqrt{1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2}} \right. \\ &\quad \left. - \frac{\left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)}{\sqrt{\left[1 - \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right)^2 \right] \left[1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2 \right]}} \right\},\end{aligned}\quad (13)$$

$$\begin{aligned}y_{32}^b &= y_1^b + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \\ &\quad \times \left\{ \sqrt{\left[1 - \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right)^2 \right]} \times \frac{1}{1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2} \right. \\ &\quad \left. + \frac{(q_1^2 - q_2^2 + d_{DE}^2)(y_2^b - y_1^b)}{2q_1 d_{DE}(x_2^a - x_1^a) \sqrt{1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2}} \right\},\end{aligned}\quad (14)$$

$$x_{33}^a = x_1^a + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \left\{ \frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE} \sqrt{1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2}} + \frac{\left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)}{\sqrt{\left[1 - \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right)^2 \right] \left[1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2 \right]}} \right\}, \quad (15)$$

$$y_{33}^b = y_1^b + q_0 \times 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} \times \left\{ \sqrt{\left[1 - \left(\frac{q_1^2 - q_2^2 + d_{DE}^2}{2q_1 d_{DE}} \right)^2 \right] \times \frac{1}{1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2}} - \frac{(q_1^2 - q_2^2 + d_{DE}^2)(y_2^b - y_1^b)}{2q_1 d_{DE}(x_2^a - x_1^a) \sqrt{1 + \left(\frac{y_2^b - y_1^b}{x_2^a - x_1^a} \right)^2}} \right\}. \quad (16)$$

However, as shown in Figure 2B, only one intersection must be selected for each pair of intersecting circles. In Figure 2C, this intersection is chosen based on the condition that the distance between the intersection point (x_{ip}, y_{ip}) and the locator node $F(x_3^a, y_3^b)$ is less than or equal to the measured distance q_3 corresponding to the locator node F . This condition is expressed as $(x_{ip} - x_3^a)^2 + (y_{ip} - y_3^b)^2 \leq q_3^2$. Finally, the intersection closer to the target can be identified. In Figure 2C, the intersection $F_1(x_{33}^a, y_{33}^b)$ satisfies the selection conditions. Similarly, the coordinates of the other two intersections $D_1(x_{11}^a, y_{11}^b)$ and $E_1(x_{22}^a, y_{22}^b)$, shown in Figure 2B, can be determined and calculated following the same steps. Once the coordinates of the three intersections shown in Figure 2B are determined, the measured distances q_1, q_2 , and q_3 are introduced as weight coefficients. Using these weights, the estimated coordinates of the target $S(x_a, y_b)$ can be calculated using the weighted formula. The EWCP algorithm is then implemented using the new weighted formulas, as shown in (17)–(19).

$$x_a = \frac{x_{11}^a \cdot q_1 + x_{22}^a \cdot q_2 + x_{33}^a \cdot q_3}{q_1 + q_2 + q_3}, y_b = \frac{y_{11}^b \cdot q_1 + y_{22}^b \cdot q_2 + y_{33}^b \cdot q_3}{q_1 + q_2 + q_3}. \quad (17)$$

$$x_a = \frac{x_{11}^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + x_{22}^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}} + x_{33}^a \cdot 10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}}}{10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}}}, \quad (18)$$

$$y_b = \frac{y_{11}^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + y_{22}^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}} + y_{33}^b \cdot 10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}}}{10^{\frac{P_t - PL(q_0) - R_{s1}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s2}}{10n}} + 10^{\frac{P_t - PL(q_0) - R_{s3}}{10n}}}. \quad (19)$$

Algorithm 1 The AIWCP Algorithm.

- 1: **Input:** $RSSI_1, RSSI_2, RSSI_3$
- 2: Calculate q_1, q_2, q_3 by (2).
- 3: Calculate the coordinates $(x_{11}^a, y_{11}^b), (x_{22}^a, y_{22}^b), (x_{33}^a, y_{33}^b)$ of the three intersections using (9)–(16).
- 4: **while** three locator nodes are mobile
- 5: Move the three locator nodes to the positions of the three intersections.
- 6: The three locator nodes transmit Wi-Fi signals, and a new set of RSSI values is obtained from the target $RSSI_{11}, RSSI_{22}, RSSI_{33}$
- 7: Calculate q_{11}, q_{22}, q_{33} using (2).
- 8: Calculate the coordinates of the three newly generated intersections using (9)–(16).
- 9: **end while**
- 10: Apply (17)–(19) to calculate the estimated coordinates of the target.

Output: The estimated coordinates (x_a, x_b)

4.2 | Adaptive iterative weighted centroid positioning algorithm

This section explains the implementation process of the AIWCP algorithm (Algorithm 1), which is specifically designed for scenarios in which the three locator nodes are movable.

The algorithm uses the proposed geometric angle method to calculate the coordinates of the intersection points and then moves the three positioning nodes to these intersection points, where they transmit Wi-Fi signals. This process generates an updated set of RSSI values and measures the distance between each locator node and the target. Subsequently, two scenarios are considered: If the locator nodes can no longer move, the algorithm directly calculates the estimated coordinates of the target. If the nodes are still movable, the process returns to the first step, recalculates the intersection coordinates based on the updated data, and adjusts the nodes accordingly.

Figure 3 illustrates one iteration of the optimization algorithm. In the left graph of Figure 3, the three locator nodes have not yet started moving. In this graph, points $D(x_1^a, y_1^b)$, $E(x_2^a, y_2^b)$, and $F(x_3^a, y_3^b)$ represent the three

locator nodes with known coordinates. Point $S(x_a, y_b)$ represent the target with unknown coordinates. The points $D_1(x_{11}^a, y_{11}^b)$, $E_1(x_{22}^a, y_{22}^b)$, and $F_1(x_{33}^a, y_{33}^b)$ represent the three intersection points whose coordinates must be calculated. The algorithm then employs the geometric angle method to determine the coordinates of these intersection points.

After determining the coordinates of the three intersections $D_1(x_{11}^a, y_{11}^b)$, $E_1(x_{22}^a, y_{22}^b)$, and $F_1(x_{33}^a, y_{33}^b)$, the three locator nodes $D(x_1^a, y_1^b)$, $E(x_2^a, y_2^b)$, and $F(x_3^a, y_3^b)$ need to be moved to the positions of these intersections, as shown in Figure 3. Subsequently, the three locator nodes emit Wi-Fi signals and a new set of RSSI values can be obtained from the target S . As a result, three newly measured distances q_{11} , q_{22} , and q_{33} between the three locator nodes and the target can be calculated using (2). At this point, three new circles are formed, resulting in the creation of three new intersection points, D_{11} , E_{11} , and F_{11} . The coordinates of these new intersection points $D_{11}(x_{04}^a, y_{04}^b)$, $E_{11}(x_{05}^a, y_{05}^b)$, and $F_{11}(x_{06}^a, y_{06}^b)$ can then be calculated by applying the proposed geometric angle method. When the three locator nodes can no

longer be moved, the estimated coordinates of the target S can be calculated using the weighted formulas (17)–(19) from the EWCP algorithm. The specific equations are provided in (20)–(23). If the three locator nodes are still movable, the algorithm proceeds to the next iteration (Figure 4).

$$x_a = \frac{x_{04}^a \times q_{11} + x_{05}^a \times q_{22} + x_{06}^a \times q_{33}}{q_{11} + q_{22} + q_{33}}, \quad (20)$$

$$y_b = \frac{y_{04}^b \times q_{11} + y_{05}^b \times q_{22} + y_{06}^b \times q_{33}}{q_{11} + q_{22} + q_{33}}. \quad (21)$$

$$x_a = \frac{x_{04}^a \cdot 10^{\frac{P_t - PL(q_0) - RS_{11}}{10n}} + x_{05}^a \cdot 10^{\frac{P_t - PL(q_0) - RS_{22}}{10n}} + x_{06}^a \cdot 10^{\frac{P_t - PL(q_0) - RS_{33}}{10n}}}{10^{\frac{P_t - PL(q_0) - RS_{11}}{10n}} + 10^{\frac{P_t - PL(q_0) - RS_{22}}{10n}} + 10^{\frac{P_t - PL(q_0) - RS_{33}}{10n}}}, \quad (22)$$

$$y_b = \frac{y_{04}^b \cdot 10^{\frac{P_t - PL(q_0) - RS_{11}}{10n}} + y_{05}^b \cdot 10^{\frac{P_t - PL(q_0) - RS_{22}}{10n}} + y_{06}^b \cdot 10^{\frac{P_t - PL(q_0) - RS_{33}}{10n}}}{10^{\frac{P_t - PL(q_0) - RS_{11}}{10n}} + 10^{\frac{P_t - PL(q_0) - RS_{22}}{10n}} + 10^{\frac{P_t - PL(q_0) - RS_{33}}{10n}}}. \quad (23)$$

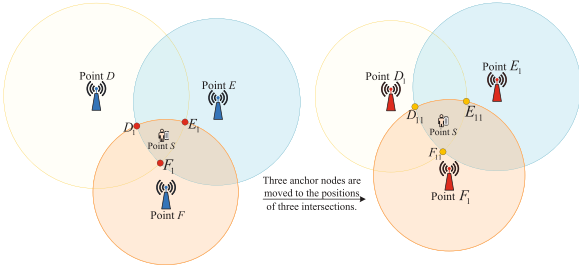


FIGURE 3 One iteration of the AIWCP algorithm.

5 | SIMULATION RESULTS

This section verifies the performance of the two proposed optimization algorithms through experimental results. The total area of the positioning scene was set to $100\text{m} \times 100\text{m}$. In the experiments, three locations were arbitrarily selected to place the three locator nodes, and one location was randomly selected to place the target.

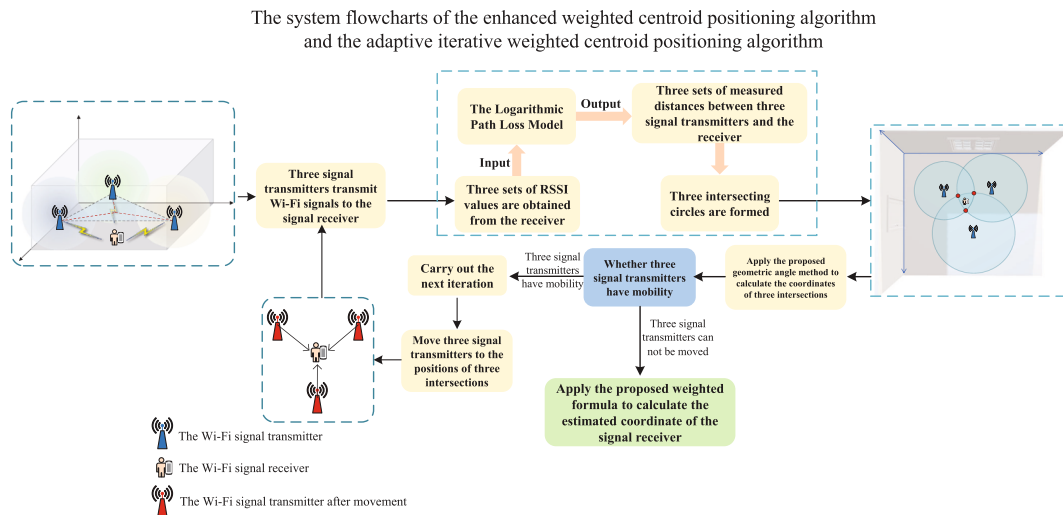


FIGURE 4 Network architecture of the proposed EWCP and AIWCP algorithms.

Subsequently, using (2), a simulated RSSI dataset was generated based on the actual distances between the three locator nodes and the target. The measured distances were then calculated according to the simulated RSSI dataset. All comparison experiments were repeated 2000 times to ensure reliability and statistical significance.

Figure 5 presents the simulation diagram of the EWCP algorithm. The positioning results of the four algorithms are shown in Figure 6, where it can be observed that the proposed EWCP algorithm achieves the highest positioning accuracy and the smallest positioning error. A more detailed comparison of the results for the five algorithms, including the CDFs of the positioning errors, is presented in Figure 7. It can be seen that the proposed EWCP algorithm has the highest CDF values, indicating the best positioning accuracy. The heat map in Figure 8 illustrates that when the positioning error was between 0 and 2 m, 82% of the experiments using the proposed EWCP algorithm achieved this accuracy. This proportion is the highest, 34% higher than that of the ITP algorithm. The stability of the five algorithms is shown in Figure 9. As seen from the boxplots, the overall positioning error values of the EWCP algorithm are the lowest, with the

error distribution being more centralized, indicating not only smaller positioning errors but also higher stability. The bar charts in Figure 10 provide a more intuitive comparison of the positioning performance of the five algorithms. The experimental results show that positioning errors of less than 2 m were the most common, accounting for the majority of 2000 experiments.

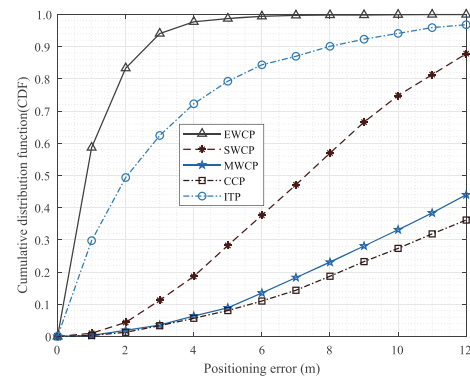


FIGURE 7 Cumulative distribution function (CDF) of positioning error for five algorithms.

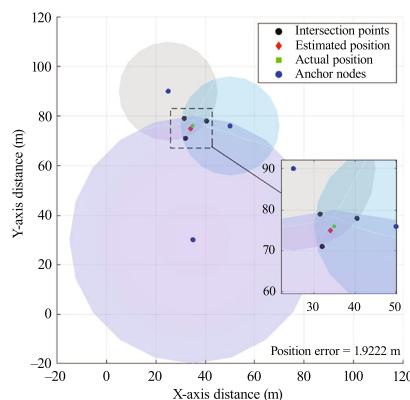


FIGURE 5 Simulation results of the EWCP algorithm.

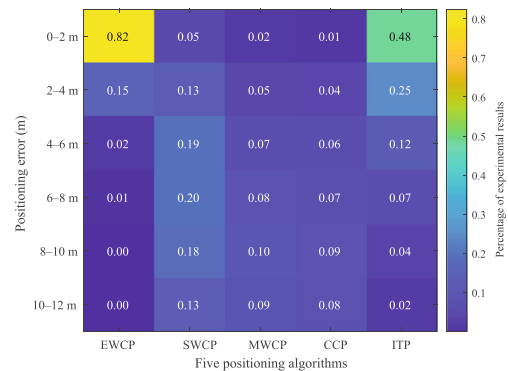


FIGURE 8 Comparison of positioning errors for five algorithms.

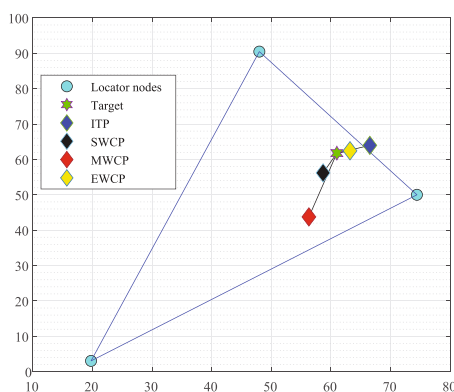


FIGURE 6 Simulation diagram of four positioning algorithms.

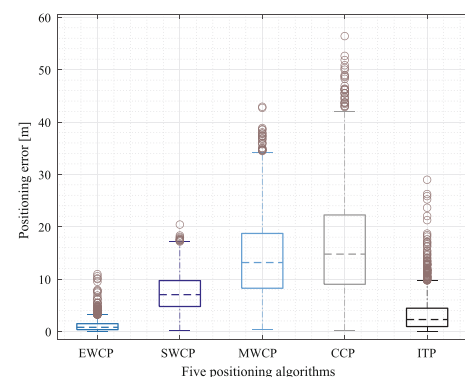


FIGURE 9 Comparison of positioning errors for five algorithms.

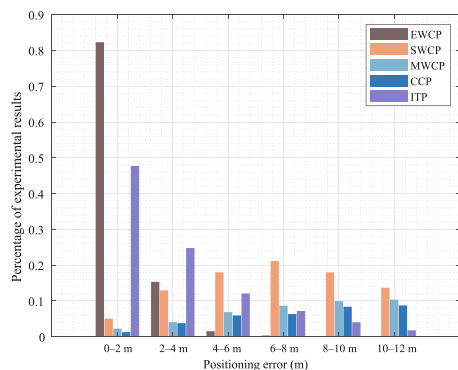


FIGURE 10 Comparison of the percentage distribution of experimental results across different positioning errors for five algorithms.

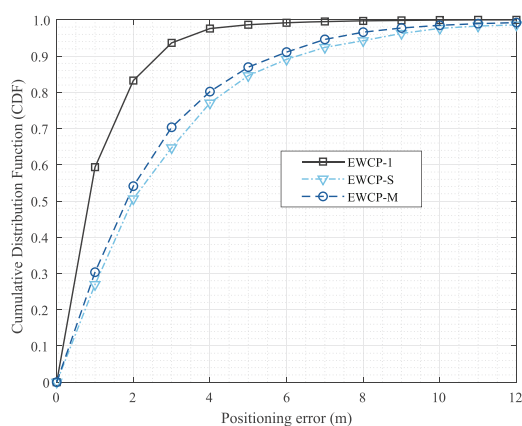


FIGURE 11 Cumulative distribution function (CDF) versus positioning errors for the EWCP algorithm with different weighted formulas.

It should be noted that the proposed EWCP algorithm applies the weighted (17)–(19) designed in this study. To further validate the positioning performance of the proposed EWCP algorithm and to demonstrate that the EWCP algorithm, when applying the proposed weighted formulas, achieves higher positioning accuracy compared to those using the weighted formulas of the MWCP and SWCP algorithms, subsequent experiments were conducted to compare these three types of weighted formulas. In Figure 11, EWCP-1 represents the EWCP algorithm as designed in this study, applying (17)–(19). EWCP-S and EWCP-M represent the EWCP algorithm using the weighted formulas from the SWCP and EWCP algorithms applying the weighted formulas of the MWCP algorithm. From the results, it can be observed that the CDF values of the EWCP algorithm applying (17)–(19) are the highest, indicating that it achieves superior positioning accuracy compared to the versions applying the weighted formulas of the MWCP and SWCP algorithms.

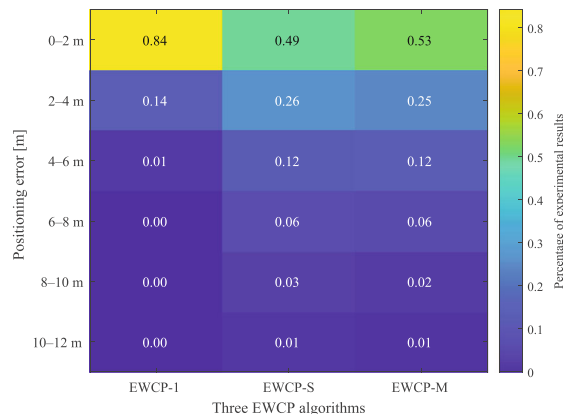


FIGURE 12 Comparison of positioning errors for three EWCP algorithms.

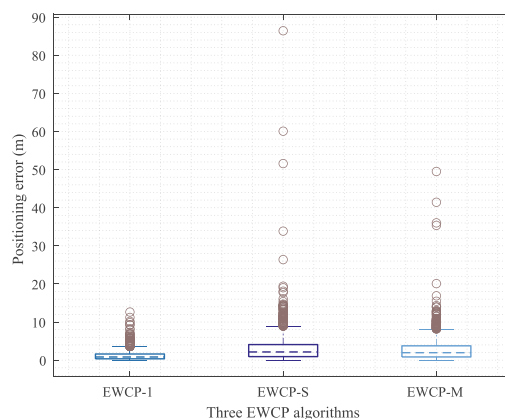


FIGURE 13 Comparison of positioning errors for three EWCP algorithms.

The specific numerical differences are shown in Figure 12. As seen in the heat maps, in the 2000 experiments, the proportion of experimental results for EWCP-1 with a positioning error of less than 2 m is 35% higher than that of EWCP-S and 31% higher than that of EWCP-M. The box plots in Figure 13 further illustrate that the positioning error for EWCP-1 has a more concentrated distribution of outliers, with the smallest average positioning error. Figure 14 visually demonstrates that, out of the 2000 experiments, EWCP-1 consistently shows better performance, particularly in terms of the number of experiments where the positioning error is between 0 and 2 m.

Figure 15 compares the AIWCP algorithm with commonly used traditional algorithms. As shown, the CDF values for AIWCP algorithm gradually increase as the number of iterations progresses. The figure also illustrates the performance of the AIWCP algorithm under the influence by interference signals in a distorted

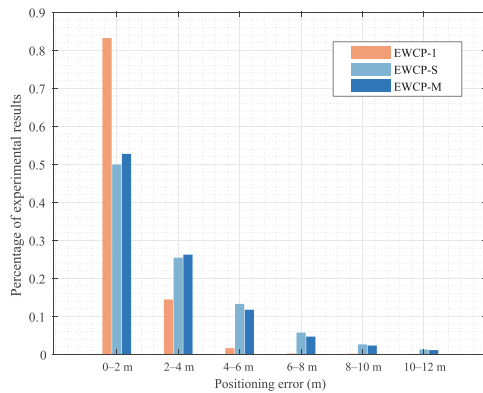


FIGURE 14 Comparison of experimental results versus positioning error for three EWCP algorithms.

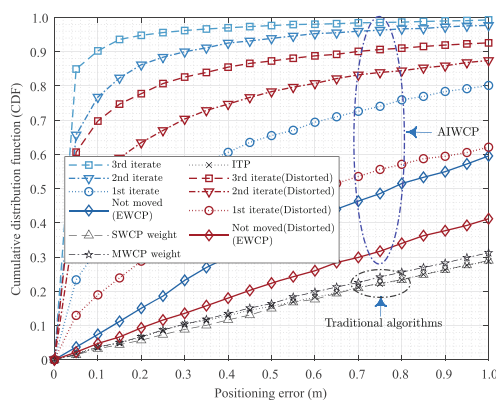


FIGURE 15 Comparison of cumulative distribution function (CDF) versus positioning error for the AIWCP algorithm and traditional algorithms.

environment, similar to an indoor space. Even in the presence of abnormal interference, the EWCP and AIWCP algorithms maintain higher positioning accuracy compared to traditional algorithms. The experimental results demonstrate that both the EWCP and AIWCP algorithms not only achieve higher positioning accuracy but also exhibit strong robustness.

The heat maps in Figure 16 show that the positioning accuracy further improves with each movement of the locator nodes. Moreover, the box plots in Figure 17 indicate that after three iterations, the distribution of positioning error values in 2000 experiments became the most concentrated. As the number of iterations increased, the positioning error decreased most significantly, as shown in Figure 18. Specifically, when the number of iterations reached three, the positioning accuracy of the AIWCP algorithm was at its highest. Following this, as the locator nodes continued to move to specified positions, the AIWCP algorithm further improved the positioning accuracy with each iteration.

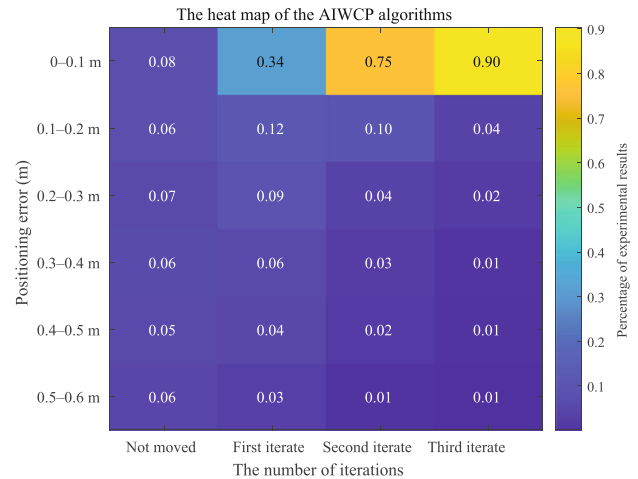


FIGURE 16 Positioning errors and the percentage of experimental results in the AIWCP algorithm.

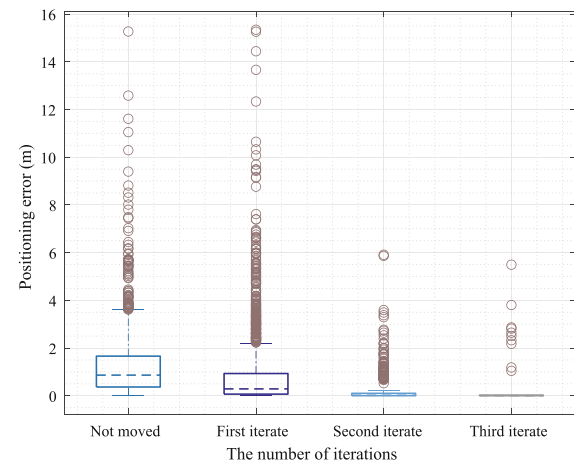


FIGURE 17 Positioning errors for the AIWCP algorithm.

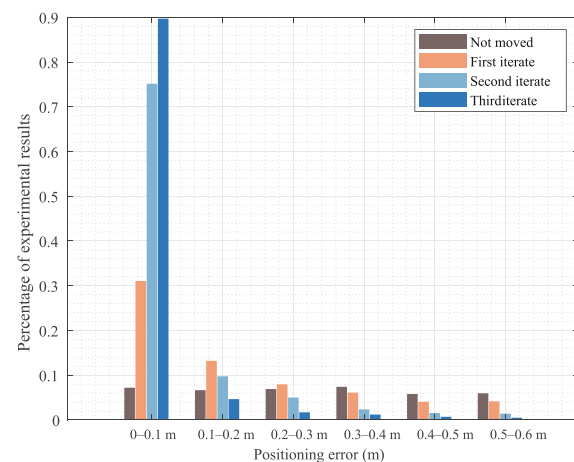


FIGURE 18 Percentage of experimental results versus positioning error for the AIWCP algorithm.

6 | CONCLUSION

This paper proposes two optimized indoor positioning algorithms based on the RSSI of Wi-Fi signals. In scenarios with only three locator nodes, the EWCP algorithm integrates the designed geometric angle method with a proposed weighting algorithm effectively leveraging the position relationships in the indoor positioning scene and the inherent properties of RSSI. The EWCP algorithm achieves high-precision indoor positioning even with a limited number of locator nodes, offering not only higher positioning accuracy but also higher resource utilization. Building upon the EWCP algorithm, the AIWCP algorithm further improves positioning accuracy by utilizing locator nodes. Each time the nodes are moved to specified positions, the AIWCP enhances accuracy. This optimization algorithm introduces an iterative approach, expanding the potential application of the weighted centroid positioning algorithm.

CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflicts of interest.

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